Mariel Del Rosario

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EDUCATION

University of California, Berkeley | GPA: 3.9/4.0 | Graduated With General Distinction **Berkeley, CA**
Bachelor of Arts in Geography, Highest Honors *May 2026*

TECHNICAL SKILLS

Geospatial Technologies: ArcGIS (Pro, Online, StoryMaps), QGIS, GDAL, Mapbox GL JS, Google Earth Pro, Georeferencing, Digitizing, Spatial Analysis (Raster & Vector), Cartography & Map Production, Coordinate Systems (CRS)
Programming and Scripting: Python (Pandas, NumPy, Matplotlib, Seaborn, GeoPandas, ArcPy), SQL (Spatial Queries), R, JavaScript, HTML5/CSS3, REST APIs
Data Analysis and Dev Tools: Shapefiles, GeoJSON, File Geodatabases, Spatial Data Cleaning, Jupyter Notebooks, Git/GitHub, Excel, Statistical Modeling (Linear Probability Models, Hypothesis Testing)

TECHNICAL PROJECTS

Web Map: Berkeley Heritage Sites (JavaScript, HTML/CSS, Mapbox) **October 2026**
- Developed an accessible, interactive web mapping application using Mapbox JS, HTML/CSS, and JavaScript to visualize historical landmark spatial data across Berkeley, CA
- Designed custom cartographic styling and responsive UI elements, including dynamic point clustering and automated attribute popups to enhance user accessibility and spatial data exploration
- Built custom JavaScript temporal filters allowing users to query spatial datasets across historical date ranges

MUNI Transit Suitability Analysis (ArcGIS Pro, Spatial Analysis) **December 2025**
- Engineered a multi-criteria spatial suitability model in ArcGIS Pro using weighted overlay methodologies to evaluate optimal public transit site locations across San Francisco
- Executed complex spatial analyses, including Euclidean distance, network analyses, Spline interpolation, and market share modeling, on multi-layered municipal datasets to motivate data-driven spatial decision-making
- Co-authored and published an interactive ArcGIS StoryMap that translates multi-step GIS methodologies into clear, accessible visual documentation and web-based applications for cross-functional stakeholders

Native American Health Equity Spatial Analysis (QGIS, Public Data) **May 2025**
- Ingested, cleaned, and normalized census and regional public health datasets to analyze socioeconomic disparities across Riverside County, CA
- Engineered 4 thematic map layers in QGIS visualizing correlations between the Health Equity Index, land use patterns, and localized health outcomes

RESEARCH EXPERIENCE

UC Berkeley Geography Department **Berkeley, CA**
Geospatial Research Assistant for Dr. Tianna Bruno *August 2024 – Present*
- Georeference historic spatial assets and locations into interactive spatial databases and digital mapping platforms
- Design spatial data repositories and collections from multiple sources, managing data organization and standardization for community-facing cartographic tools
- Co-authored and presented spatial environmental justice research at the 2026 AAG Annual Conference
- Collaborated in a 4-person team to build interactive digital storymaps documenting regional environmental injustice and historical geography

WORK EXPERIENCE

College of Computing, Data Science, and Society at UC Berkeley **Berkeley, CA**
Lead Teaching Assistant and Senior Mentor for Foundations of Data Science *August 2023 – May 2026*
- Automated backend workflows for 34 staff members using Google Colab and Google Sheets integrations
- Co-authored, tested, and debugged Python programming assignments and labs for 1,500+ students
- Supervised 40+ TAs/tutors on technical pedagogy, Python debugging, and statistical inference

Pilipinx Academic Student Services **Berkeley, CA**
Retention Director and Gender/Sexuality Awareness Coordinator *May 2023 – June 2025*
- Developed custom spatial databases via Google Maps to optimize vendor routing, venue selection, and logistical event planning for large-scale operations
- Managed a $26,000+ budget and led a 15-person team organizing student retention initiatives